1. What is an Operating System, and list types of Operating Systems [5M]

An Operating System (OS) is system software that acts as an interface between the user and computer hardware. It manages hardware resources, controls execution of programs, and provides services for application software.

The operating system is responsible for:

- Process management
- Memory management
- File management
- Device management
- Security and protection
- CPU scheduling

Without an operating system, users cannot effectively interact with the computer system.

Types of Operating Systems

1. Batch Operating System

Executes jobs in batches without direct user interaction.

2. Time Sharing Operating System

Allows multiple users to access the system simultaneously using time slicing.

3. Real-Time Operating System (RTOS)

Provides immediate response for time-critical applications.

4. Distributed Operating System

Manages multiple computers and makes them work as a single system.

5. Network Operating System

Controls and manages network resources and communication.

6. Multiprocessing Operating System

Uses multiple processors for parallel execution.

7. Mobile Operating System

Designed for mobile devices such as Android and iOS.

Conclusion

The operating system is an essential component of a computer system because it manages resources and enables smooth interaction between users and hardware.

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2. What is a clock in a CPU? Explain over-clocking [5M]

The clock in a CPU is an electronic timing signal that synchronizes all operations performed inside the processor. It determines how fast instructions are executed.

The CPU performs operations according to clock cycles generated by the clock signal. Clock speed is measured in Hertz (Hz), mainly in Gigahertz (GHz).

For example:

- 1 GHz = 1 billion cycles per second

A higher clock speed generally means faster instruction execution.

Over-clocking

Over-clocking is the process of increasing the clock speed of the processor beyond the manufacturer's specified limit to improve performance.

It is commonly used in:

- Gaming
- Video editing

- Simulations
- High-performance computing

Advantages of Over-clocking

1. Improved processing speed
2. Better gaming performance
3. Faster execution of applications

Disadvantages of Over-clocking

1. Increased heat generation
2. Higher power consumption
3. System instability
4. Reduced hardware lifespan

Proper cooling systems are required for safe over-clocking.

Conclusion

The CPU clock controls processor operations, while over-clocking increases performance at the cost of heat and power consumption.

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3. Explain how DeadLock occurs, and tell its prevention methods [10M]

Deadlock is a condition in which two or more processes are unable to continue execution because each process is waiting for resources held by another process.

In a deadlock state, processes remain blocked indefinitely, resulting in poor resource utilization and reduced system performance.

Example of Deadlock

Suppose:

- Process P1 holds Resource R1 and waits for Resource R2.
- Process P2 holds Resource R2 and waits for Resource R1.

Both processes wait forever, causing deadlock.

Necessary Conditions for Deadlock

According to Coffman's conditions, deadlock occurs only when all four conditions are present simultaneously.

1. Mutual Exclusion

A resource can be used by only one process at a time.

2. Hold and Wait

A process holding resources waits for additional resources.

3. No Preemption

Resources cannot be forcibly removed from a process.

4. Circular Wait

Processes form a circular chain where each process waits for a resource held by another process.

Prevention Methods

1. Eliminate Mutual Exclusion

Allow sharing of resources whenever possible.

2. Eliminate Hold and Wait

Processes must request all resources before execution begins.

3. Eliminate No Preemption

Resources may be taken back from processes if necessary.

4. Eliminate Circular Wait

Resources should be requested in a fixed order.

Deadlock Avoidance

Deadlock avoidance ensures the system never enters an unsafe state.

Banker's Algorithm

A famous deadlock avoidance algorithm that checks resource allocation safety before granting requests.

Deadlock Detection and Recovery

Some systems allow deadlock occurrence and later:

- Detect deadlock
- Recover by terminating processes or reallocating resources

Effects of Deadlock

- CPU underutilization
- Process starvation
- Resource wastage
- Reduced system efficiency

Conclusion

Deadlock is a serious problem in operating systems that can completely stop process execution. Proper resource allocation strategies and prevention techniques are essential for stable system operation.

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4. What is a Thread? Explain Multi-Threading [10M]

A thread is the smallest unit of CPU execution within a process. A process may contain multiple threads that execute concurrently while sharing the same memory and resources.

Each thread has:

- Program counter
- Registers
- Stack

Threads are lightweight compared to processes because they share resources.

Multi-Threading

Multi-threading is the capability of executing multiple threads simultaneously within a single process.

It improves performance, responsiveness, and CPU utilization.

Example

In a web browser:

- One thread loads webpages
- One thread downloads files
- One thread handles user input

All tasks execute simultaneously.

Types of Threads

1. User-Level Threads

Managed by user-level libraries.

2. Kernel-Level Threads

Managed directly by the operating system kernel.

Advantages of Multi-Threading

1. Improved Responsiveness

Applications remain active during background tasks.

2. Better CPU Utilization

Multiple threads execute efficiently on multi-core systems.

3. Faster Execution

Tasks execute concurrently.

4. Resource Sharing

Threads share memory and files.

5. Economical

Thread creation requires less overhead than process creation.

Disadvantages of Multi-Threading

- Synchronization problems
- Race conditions
- Deadlocks
- Difficult debugging

Applications of Multi-Threading

- Operating systems
- Web servers
- Gaming engines
- Multimedia software
- Database systems

Conclusion

Multi-threading is an important concept in modern operating systems because it increases efficiency, responsiveness, and overall system performance.

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5. Explain FCFS, Round Robin, SJF Algorithms and which is Optimal. Justify your answer. [10M]

CPU scheduling algorithms determine the order in which processes receive CPU time.

Efficient scheduling improves:

- CPU utilization
- Throughput
- Waiting time
- Response time

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1. FCFS (First Come First Serve)

In FCFS scheduling, the process that arrives first gets executed first.

It follows FIFO order.

Advantages

- Simple implementation
- Fair according to arrival order

Disadvantages

- High waiting time
- Convoy effect
- Poor responsiveness

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2. Round Robin Scheduling

In Round Robin scheduling, each process gets CPU time for a fixed time quantum.

After the quantum expires, the process moves to the back of the queue.

Advantages

- Fair CPU allocation
- Good response time

- Suitable for time-sharing systems

Disadvantages

- Frequent context switching
- Performance depends on time quantum

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3. Shortest Job First (SJF)

In SJF scheduling, the process with the shortest burst time executes first.

Advantages

- Minimum average waiting time
- High throughput
- Efficient CPU utilization

Disadvantages

- Starvation of long processes
- Difficult burst time prediction

Which is Optimal?

SJF is considered the optimal scheduling algorithm because it mathematically produces the minimum average waiting time among scheduling algorithms.

Short processes finish quickly, reducing overall waiting time and improving throughput.

However:

- Round Robin is more suitable for interactive systems.
- FCFS is simple but less efficient.

Conclusion

Among FCFS, Round Robin, and SJF, SJF is theoretically optimal for minimizing waiting time, while Round Robin is more practical for multitasking systems.

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6. Explain Paging and Segmentation. What are the differences [10M]

Paging and segmentation are memory management techniques used by operating systems to allocate memory efficiently.

Paging

Paging divides:

- Logical memory into fixed-size pages
- Physical memory into fixed-size frames

Pages are mapped to frames using a page table.

Advantages of Paging

1. Eliminates external fragmentation
2. Efficient memory utilization
3. Supports virtual memory

Disadvantages of Paging

1. Internal fragmentation
2. Page table overhead

Segmentation

Segmentation divides memory into variable-sized logical units called segments.

Examples:

- Code segment
- Data segment
- Stack segment

Each segment contains:

- Base address
- Limit value

Advantages of Segmentation

1. Logical program organization
2. Easy protection and sharing
3. Better modularity

Disadvantages of Segmentation

1. External fragmentation
2. Complex memory allocation

Differences Between Paging and Segmentation

Paging

- Fixed-size blocks
- Physical memory division
- Invisible to programmer
- Internal fragmentation occurs

Segmentation

- Variable-size blocks
- Logical memory division
- Visible to programmer
- External fragmentation occurs

Conclusion

Paging improves efficient memory allocation, while segmentation provides logical organization of programs. Modern operating systems often combine both methods for effective memory management.